

Linux Class 12 - Practice Task

Practising find by size and depth, and the network commands

Submitted by: Shivakumar Muradi

Date: 6 October 2026

find by size - the letters

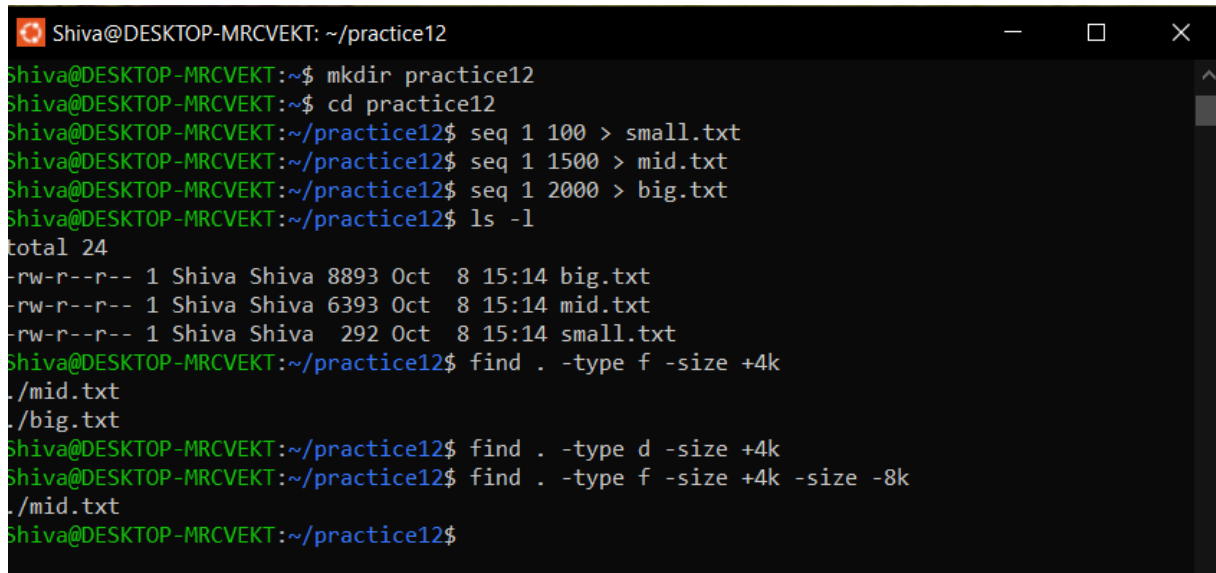
From the notes, -size takes a number and a letter. + means bigger than and - means smaller than.

Letter	Means	Example from the notes
c	bytes	-size +20c bigger than 20 bytes
k	kilobytes (1 KB = 1024 bytes)	-size +10k bigger than 10 KB
M	megabytes	-size +10M bigger than 10 MB
G	gigabytes	-size -2G smaller than 2 GB
b	blocks of 512 bytes	-size +4b bigger than $4 \times 512 = 2048$ bytes (2 KB)
w	words of 2 bytes	-size +100w bigger than $100 \times 2 = 200$ bytes

Making files of different sizes

To test sizes I needed files that are big and small. I used seq, which just counts numbers into a file, so I could make three files of different sizes quickly.

```
seq 1 100 > small.txt
seq 1 1500 > mid.txt
seq 1 2000 > big.txt
ls -l
find . -type f -size +4k
find . -type d -size +4k
find . -type f -size +4k -size -8k
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~$ mkdir practice12
Shiva@DESKTOP-MRCVEKT:~$ cd practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ seq 1 100 > small.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ seq 1 1500 > mid.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ seq 1 2000 > big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ ls -l
total 24
-rw-r--r-- 1 Shiva Shiva 8893 Oct  8 15:14 big.txt
-rw-r--r-- 1 Shiva Shiva 6393 Oct  8 15:14 mid.txt
-rw-r--r-- 1 Shiva Shiva  292 Oct  8 15:14 small.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +4k
./mid.txt
./big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type d -size +4k
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +4k -size -8k
./mid.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

Screenshot 1: three files, then the three find commands from the notes.

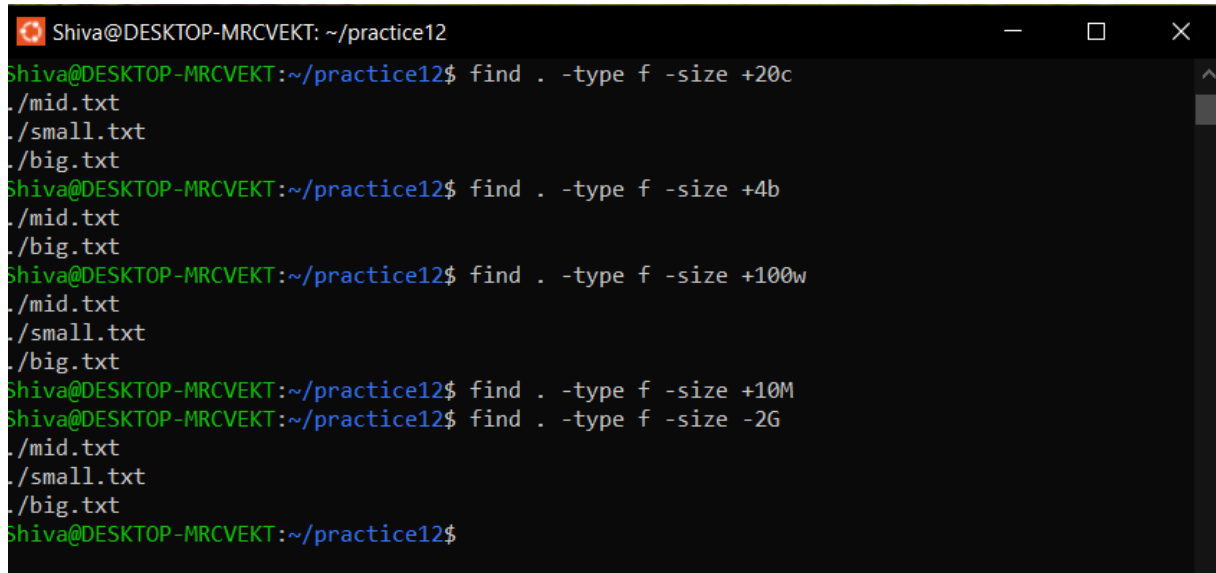
ls -l shows big.txt is 8893 bytes (about 8.7 KB), mid.txt is 6393 bytes (about 6.2 KB) and small.txt is only 292 bytes.

- -size +4k gave mid.txt and big.txt. That is right, both are more than 4 KB.
- -type d -size +4k gave nothing. I was confused at first, but a folder in Linux is exactly 4096 bytes, which is 4 KB and not more than 4 KB. So there was no folder bigger than 4k.
- -size +4k -size -8k gave only mid.txt. Two -size in one command means both must be true, so it is like between 4 KB and 8 KB. big.txt is more than 8 KB so it was left out.

One small thing I noticed: find rounds the size up to full KB. So mid.txt (6.2 KB) counts as 7k and big.txt (8.7 KB) counts as 9k. That is why big.txt is not less than 8k.

Trying the other size letters

```
find . -type f -size +20c
find . -type f -size +4b
find . -type f -size +100w
find . -type f -size +10M
find . -type f -size -2G
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +20c
./mid.txt
./small.txt
./big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +4b
./mid.txt
./big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +100w
./mid.txt
./small.txt
./big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size +10M
Shiva@DESKTOP-MRCVEKT:~/practice12$ find . -type f -size -2G
./mid.txt
./small.txt
./big.txt
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

Screenshot 2: c, b, w, M and G on the same three files.

+20c (more than 20 bytes) gave all three files, because even small.txt is 292 bytes.

+4b means more than 4 blocks of 512 bytes, which is 2048 bytes. small.txt is not that big, so only mid.txt and big.txt came.

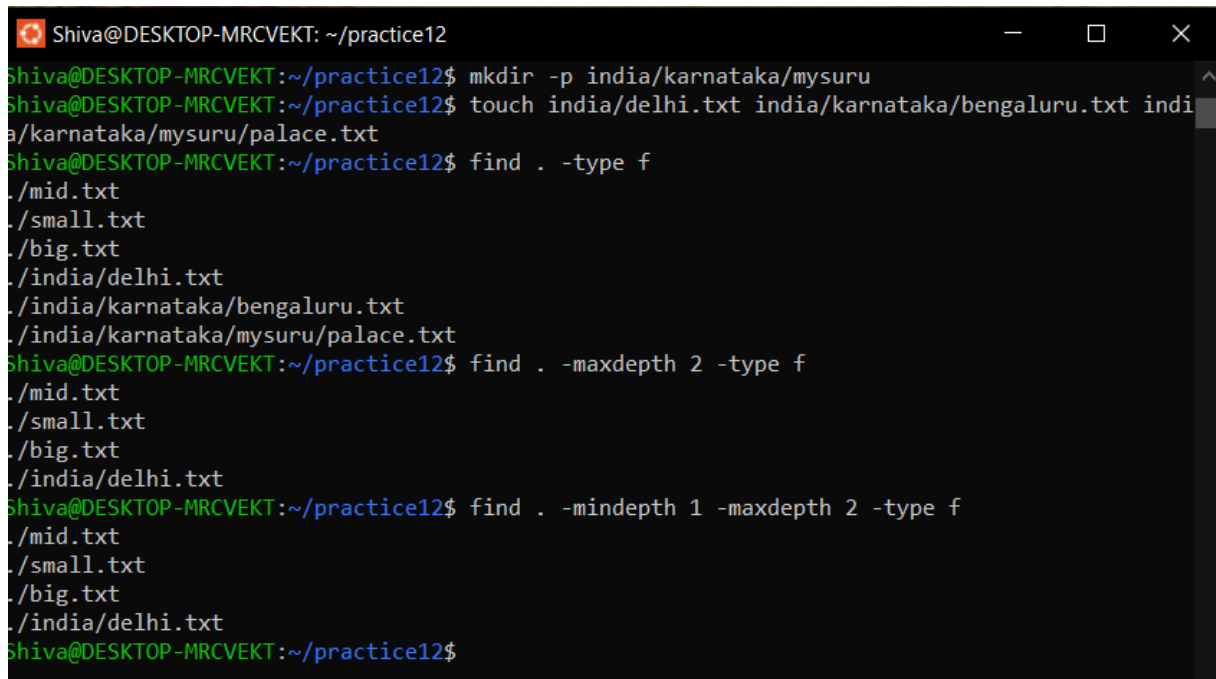
+100w means more than 100 words of 2 bytes, which is 200 bytes. small.txt is 292 bytes, so this time all three came.

+10M gave nothing because none of my files is that big, and -2G gave all three because every file is smaller than 2 GB.

Searching only a few levels deep

For this I made folders inside folders: india, then karnataka inside it, then mysuru inside that. -p in mkdir makes all of them in one go. Then I put a file at every level.

```
mkdir -p india/karnataka/mysuru  
touch india/delhi.txt india/karnataka/bengaluru.txt  
india/karnataka/mysuru/palace.txt  
find . -type f  
find . -maxdepth 2 -type f  
find . -mindepth 1 -maxdepth 2 -type f
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12'. The terminal shows the following commands and their outputs:
1. Command: `mkdir -p india/karnataka/mysuru`
2. Command: `touch india/delhi.txt india/karnataka/bengaluru.txt india/karnataka/mysuru/palace.txt`
3. Command: `find . -type f`
Output: `./mid.txt
./small.txt
./big.txt
./india/delhi.txt
./india/karnataka/bengaluru.txt
./india/karnataka/mysuru/palace.txt`
4. Command: `find . -maxdepth 2 -type f`
Output: `./mid.txt
./small.txt
./big.txt
./india/delhi.txt`
5. Command: `find . -mindepth 1 -maxdepth 2 -type f`
Output: `./mid.txt
./small.txt
./big.txt
./india/delhi.txt`
The terminal prompt is `Shiva@DESKTOP-MRCVEKT:~/practice12$`.

Screenshot 3: all files, then only up to 2 levels deep.

Plain `find . -type f` went into every folder and showed all 6 files.

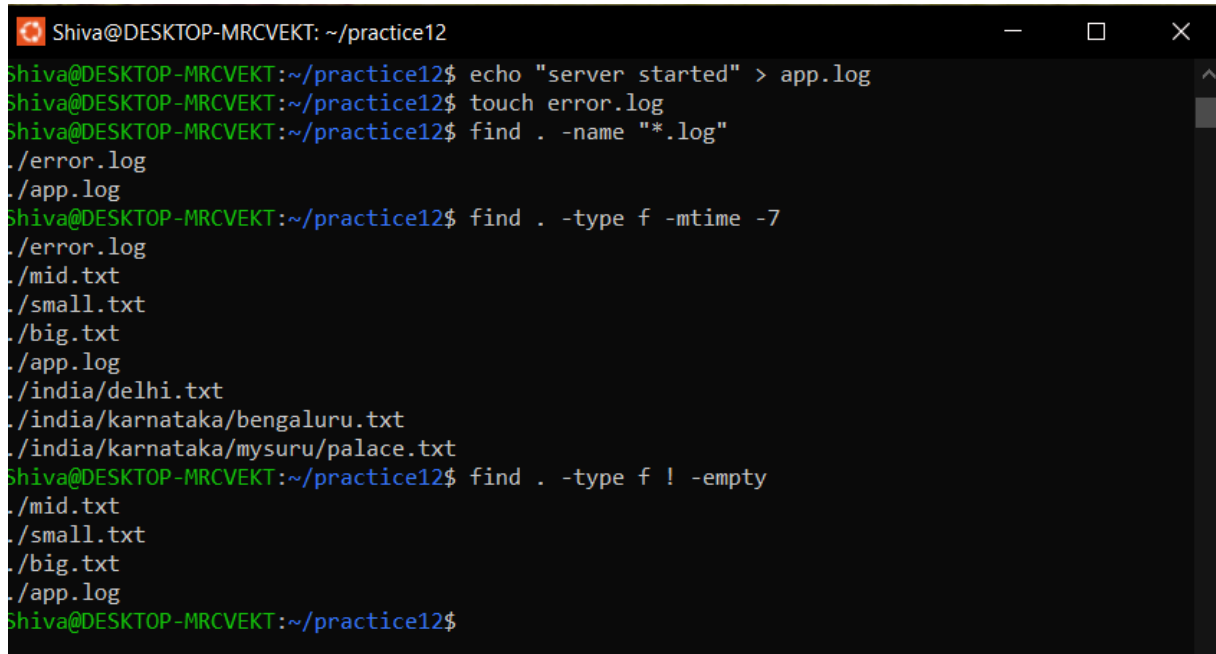
`-maxdepth 2` stopped after 2 levels. Level 1 is the files right here (small, mid, big) and level 2 is `india/delhi.txt`. `bengaluru.txt` is at level 3 and `palace.txt` is at level 4, so they were not shown.

`-mindepth 1 -maxdepth 2` gave the same list. That is because level 0 is the current folder (.) itself, and it is a folder, not a file. So leaving it out made no difference here with `-type f`.

Finding by name, by date, and files that are not empty

I made two `.log` files, one with text and one empty, so I could try the last three `find` commands from the notes.

```
echo "server started" > app.log
touch error.log
find . -name "*.log"
find . -type f -mtime -7
find . -type f ! -empty
```

A screenshot of a terminal window titled "Shiva@DESKTOP-MRCVEKT: ~/practice12". The terminal shows the following commands and their outputs:
1. Command: `echo "server started" > app.log`
2. Command: `touch error.log`
3. Command: `find . -name "*.log"`
Output: `./error.log`
`./app.log`
4. Command: `find . -type f -mtime -7`
Output: `./error.log`
`./mid.txt`
`./small.txt`
`./big.txt`
`./app.log`
`./india/delhi.txt`
`./india/karnataka/bengaluru.txt`
`./india/karnataka/mysuru/palace.txt`
5. Command: `find . -type f ! -empty`
Output: `./mid.txt`
`./small.txt`
`./big.txt`
`./app.log`
The prompt `Shiva@DESKTOP-MRCVEKT:~/practice12$` is visible at the end of the last command.

Screenshot 4: .log files, files changed in the last 7 days, and files that are not empty.

`-name "*.log"` found `app.log` and `error.log`. The `*` means anything, so any name ending with `.log`. The quotes are needed so the shell does not change the `*` by itself.

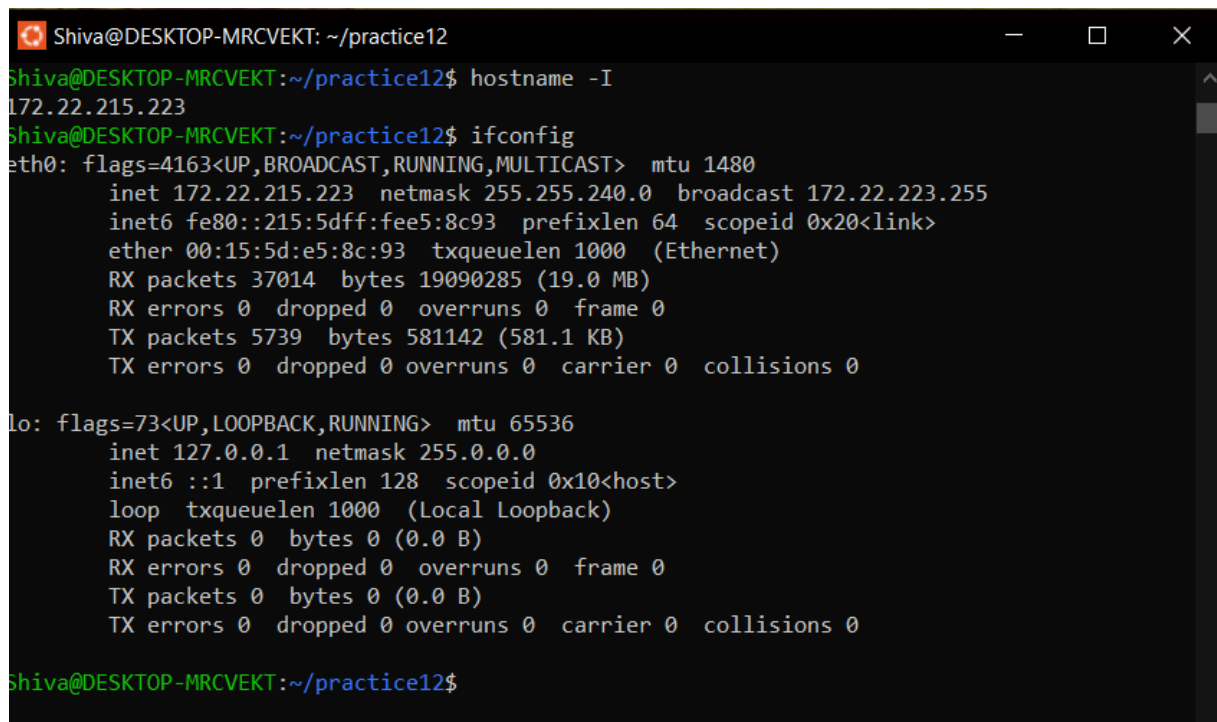
`-mtime -7` means changed in the last 7 days. I made all these files today, so all of them came.

`!` means not, so `! -empty` is files that are not empty. It showed only `mid`, `small`, `big` and `app.log`. `error.log` and the 3 files in the `india` folders are empty because I made them with `touch`.

Network commands - my IP address

```
hostname -I
```

```
ifconfig
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12'. The terminal shows the following commands and their outputs:
1. Command: `hostname -I`
Output: `172.22.215.223`
2. Command: `ifconfig`
Output:
`eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1480`
`inet 172.22.215.223 netmask 255.255.240.0 broadcast 172.22.223.255`
`inet6 fe80::215:5dff:fee5:8c93 prefixlen 64 scopeid 0x20<link>`
`ether 00:15:5d:e5:8c:93 txqueuelen 1000 (Ethernet)`
`RX packets 37014 bytes 19090285 (19.0 MB)`
`RX errors 0 dropped 0 overruns 0 frame 0`
`TX packets 5739 bytes 581142 (581.1 KB)`
`TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0`

`lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536`
`inet 127.0.0.1 netmask 255.0.0.0`
`inet6 ::1 prefixlen 128 scopeid 0x10<host>`
`loop txqueuelen 1000 (Local Loopback)`
`RX packets 0 bytes 0 (0.0 B)`
`RX errors 0 dropped 0 overruns 0 frame 0`
`TX packets 0 bytes 0 (0.0 B)`
`TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0`

The prompt `Shiva@DESKTOP-MRCVEKT:~/practice12$` is visible at the bottom.

Screenshot 5: `hostname -I` gives just the IP, `ifconfig` gives the full details.

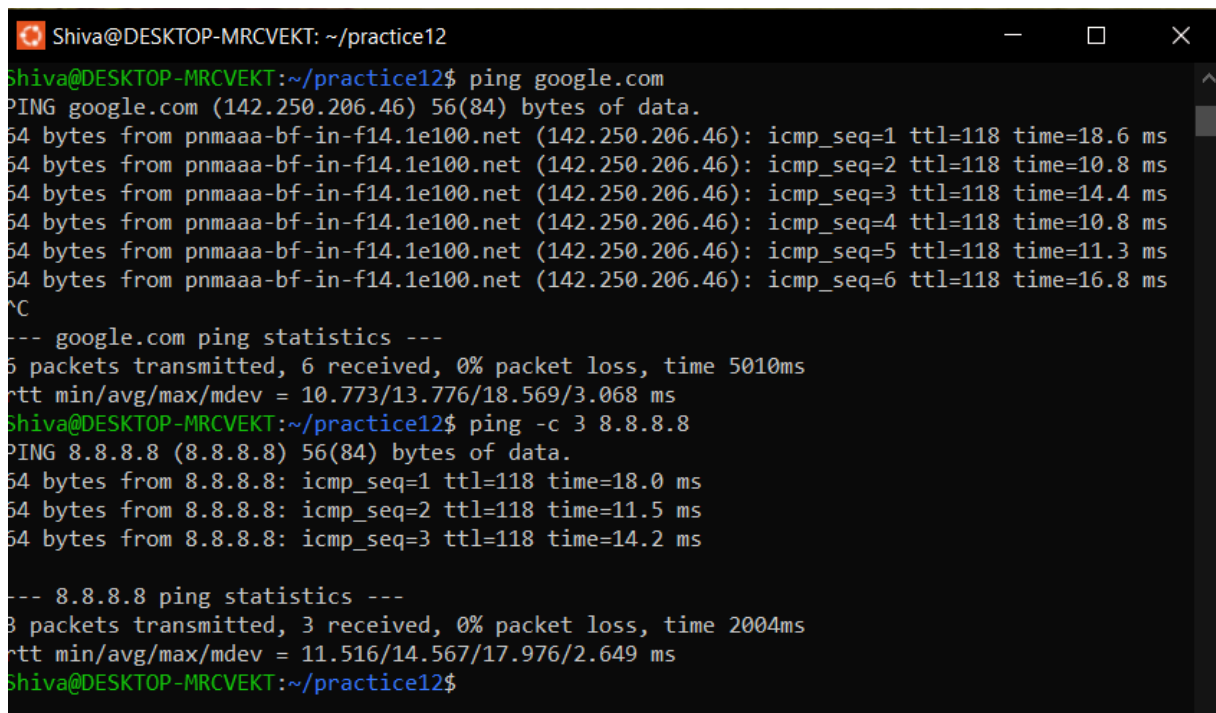
`hostname -I` just printed my IP address, 172.22.215.223. `ifconfig` showed the same address on the line starting with `inet`, plus a lot of other details like netmask and how many packets went in (RX) and out (TX).

The notes say `eno1` is the ethernet cable, `wlo1` is wifi and `lo` is the computer talking to itself. On my computer I see `eth0` and `lo` instead. I think it is because my Ubuntu runs inside Windows (WSL), so Ubuntu gets one network card called `eth0` from Windows, and Windows handles the real wifi. `lo` has 127.0.0.1, which is the computer talking to itself, same as the notes say.

ping

`ping` checks if another computer is reachable. The notes show `ping google.com`. On Linux this keeps going forever, so I stopped it with `Ctrl+C`. Then I tried `ping` with an IP address and `-c 3`, which stops by itself after 3 tries.

```
ping google.com
ping -c 3 8.8.8.8
```

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12' showing the output of two ping commands. The first command is 'ping google.com', which shows six successful pings from 'pnmaaa-bf-in-f14.1e100.net' (142.250.206.46) with varying times (18.6ms to 16.8ms). The second command is 'ping -c 3 8.8.8.8', showing three successful pings from '8.8.8.8' with times of 18.0ms, 11.5ms, and 14.2ms. Both commands show 0% packet loss and a summary of statistics at the end.

```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ ping google.com
PING google.com (142.250.206.46) 56(84) bytes of data.
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=1 ttl=118 time=18.6 ms
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=2 ttl=118 time=10.8 ms
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=3 ttl=118 time=14.4 ms
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=4 ttl=118 time=10.8 ms
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=5 ttl=118 time=11.3 ms
64 bytes from pnmaaa-bf-in-f14.1e100.net (142.250.206.46): icmp_seq=6 ttl=118 time=16.8 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 10.773/13.776/18.569/3.068 ms
Shiva@DESKTOP-MRCVEKT:~/practice12$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=18.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=11.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=14.2 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 11.516/14.567/17.976/2.649 ms
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

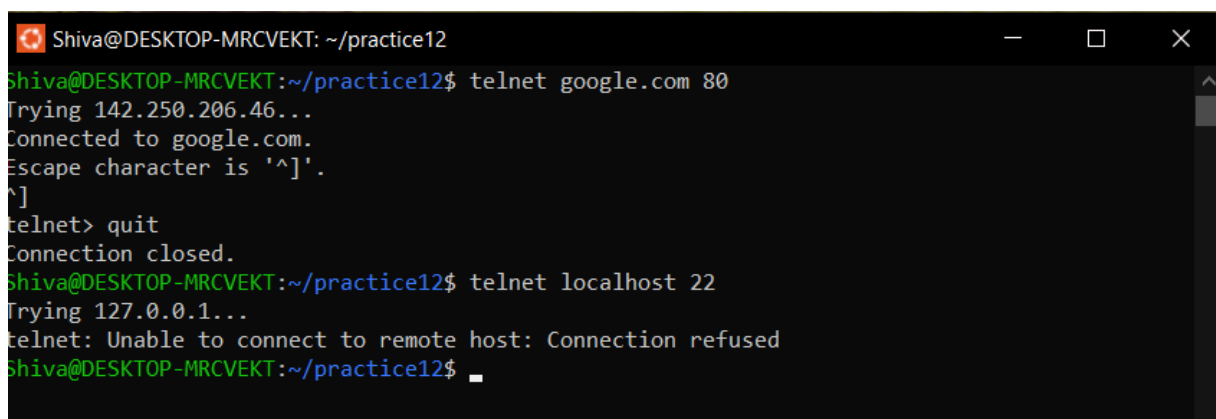
Screenshot 6: ping to a name and to an IP address. 0% packet loss both times.

Every line is one answer from the other computer, and time= shows how many milliseconds it took. At the end it shows a summary: 6 sent, 6 received, 0% packet loss. So google.com is reachable. 8.8.8.8 is Google's DNS server, I used it to try ping with an IP address like in the notes.

telnet

telnet tests if one port on a computer is open. The example in the notes is google.com on port 80, which is http. I also tried port 22 (ssh) on my own computer, to see what a closed port looks like.

```
telnet google.com 80
telnet localhost 22
```

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12' showing the output of two telnet commands. The first command is 'telnet google.com 80', which shows a successful connection to '142.250.206.46' on port 80. The user types 'quit' and the connection is closed. The second command is 'telnet localhost 22', which shows a failed connection to '127.0.0.1' on port 22 with the message 'Connection refused'.

```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ telnet google.com 80
Trying 142.250.206.46...
Connected to google.com.
Escape character is '^['.
^]
telnet> quit
Connection closed.
Shiva@DESKTOP-MRCVEKT:~/practice12$ telnet localhost 22
Trying 127.0.0.1...
telnet: Unable to connect to remote host: Connection refused
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

Screenshot 7: port 80 on google.com is open, port 22 on my computer is closed.

For google.com it said Connected to google.com, so port 80 is open. After that it waits for me to type, so to come out I pressed Ctrl+] which gives a telnet> prompt, and typed quit.

For localhost 22 it said Connection refused. That means nothing is listening on port 22 on my computer, because I do not have an ssh server running.

The ports from the notes:

Service	Port
ssh	22
telnet	23
smtp (mail)	25
dns	53
http	80
https	443

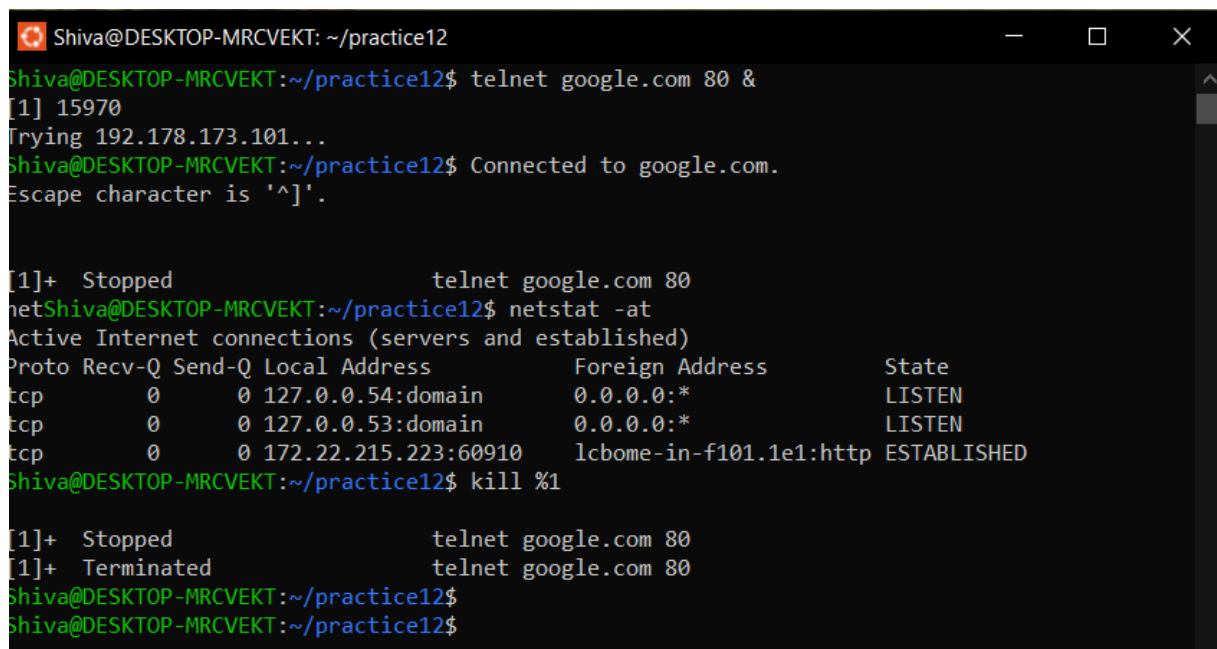
netstat

netstat shows the network connections on my own computer. I wanted to see an ESTABLISHED connection, so I started telnet to google.com in the background with & at the end. Then I ran netstat -at, which shows only the TCP connections, and then stopped the telnet with kill %1.

```
telnet google.com 80 &
```

```
netstat -at
```

```
kill %1
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12'. The terminal shows the following sequence of commands and output:
1. `telnet google.com 80 &`
Output: `[1] 15970`
`Trying 192.178.173.101...`
`Shiva@DESKTOP-MRCVEKT:~/practice12$ Connected to google.com.`
`Escape character is '^]'.`
2. `netstat -at`
Output: `[1]+ Stopped telnet google.com 80`
`Shiva@DESKTOP-MRCVEKT:~/practice12$ netstat -at`
Output: `Active Internet connections (servers and established)`

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	127.0.0.54:domain	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.53:domain	0.0.0.0:*	LISTEN
tcp	0	0	172.22.215.223:60910	lcbome-in-f101.1e1:http	ESTABLISHED

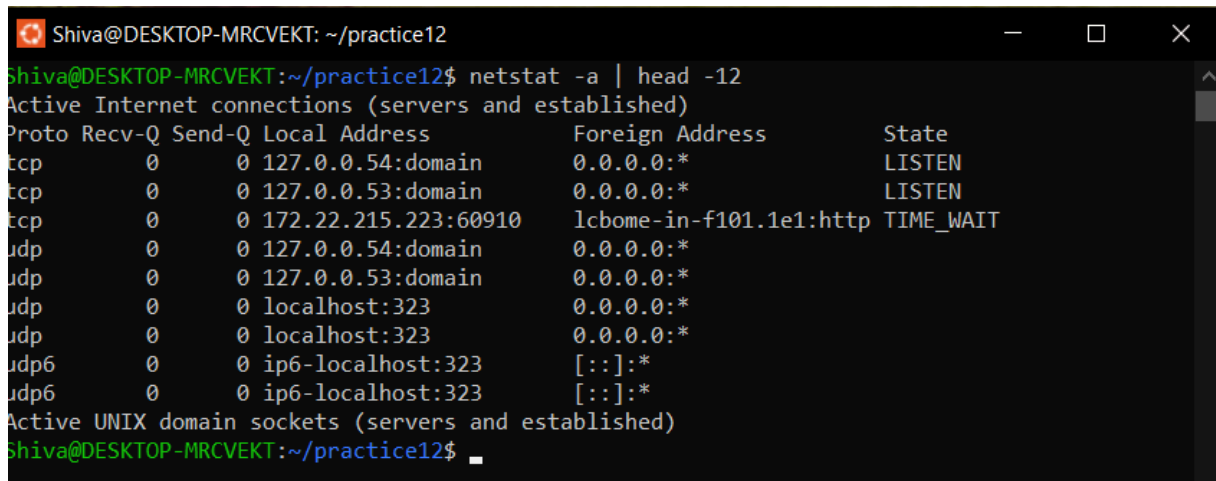
3. `kill %1`
Output: `[1]+ Stopped telnet google.com 80`
`[1]+ Terminated telnet google.com 80`
`Shiva@DESKTOP-MRCVEKT:~/practice12$`
`Shiva@DESKTOP-MRCVEKT:~/practice12$`

Screenshot 8: two LISTEN lines and one ESTABLISHED line to google.com.

The output is a bit messy at the top because telnet printed its messages in the middle of my prompt, but netstat worked. The two LISTEN lines are on port domain (53, which is dns). The third line is my telnet: from my IP to google on http, and the state is ESTABLISHED, which means the connection is active.

Then the command from the notes, netstat -a. It prints a very long list, so I added | head -12 to see only the first 12 lines.

```
netstat -a | head -12
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ netstat -a | head -12
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.54:domain       0.0.0.0:*              LISTEN
tcp        0      0 127.0.0.53:domain       0.0.0.0:*              LISTEN
tcp        0      0 172.22.215.223:60910    lcbome-in-f101.1e1:http TIME_WAIT
udp        0      0 127.0.0.54:domain       0.0.0.0:*
udp        0      0 127.0.0.53:domain       0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 ip6-localhost:323      [::]:*
udp6       0      0 ip6-localhost:323      [::]:*
Active UNIX domain sockets (servers and established)
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

Screenshot 9: netstat -a shows tcp and udp, and the google line is now TIME_WAIT.

The line to google now says TIME_WAIT instead of ESTABLISHED, because I had just closed that telnet. It means the connection is finished and the computer waits a little before forgetting it. -a also shows udp lines, which have no state.

The notes list three states: OPEN, LISTEN and ESTABLISHED. I saw LISTEN and ESTABLISHED in my output. I did not see a state called OPEN in netstat; in my output an open port shows up as LISTEN.

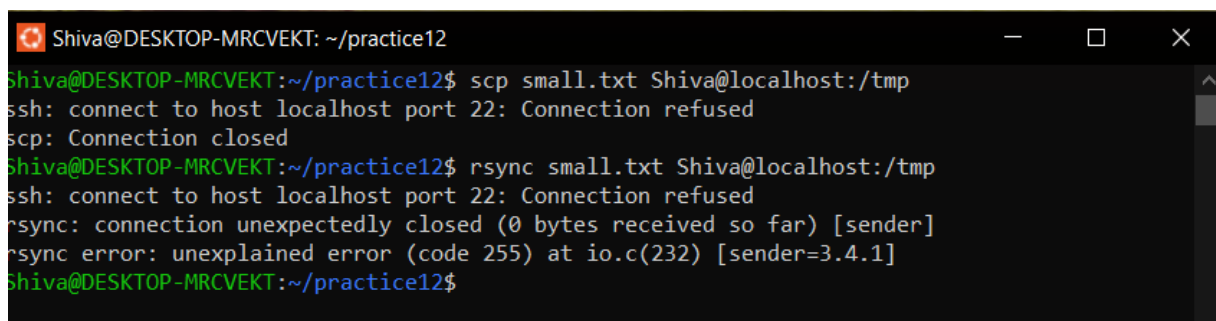
scp and rsync

Both are for copying files between my computer and a server. The syntax from the notes is:

```
scp <file-path> <username>@<server-ip>:<server-path>
scp <username>@<server-ip>:<server-path> <local-path>
rsync <file-path> <username>@<server-ip>:<server-path>
rsync <username>@<server-ip>:<server-path> <local-path>
```

I do not have a remote server, so I tried copying to my own computer as if it was a server, using localhost.

```
scp small.txt Shiva@localhost:/tmp
rsync small.txt Shiva@localhost:/tmp
```



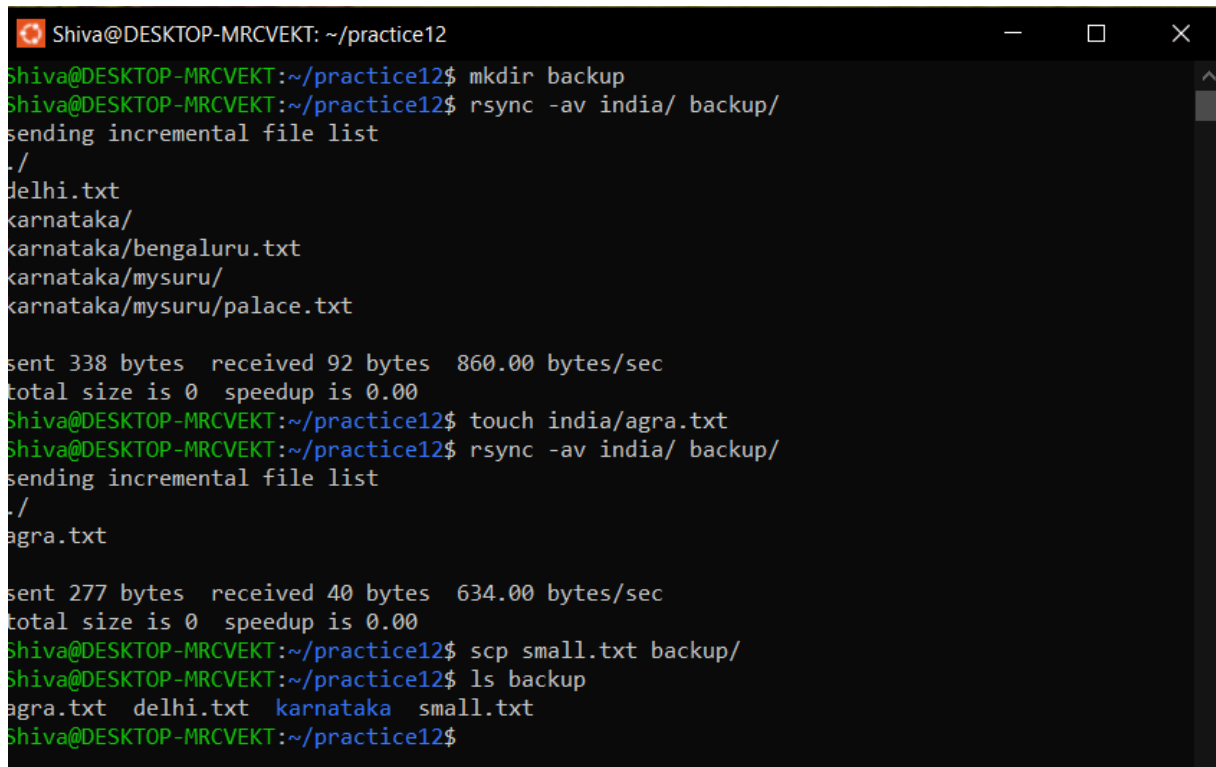
```
Shiva@DESKTOP-MRCVEKT: ~/practice12
Shiva@DESKTOP-MRCVEKT:~/practice12$ scp small.txt Shiva@localhost:/tmp
ssh: connect to host localhost port 22: Connection refused
scp: Connection closed
Shiva@DESKTOP-MRCVEKT:~/practice12$ rsync small.txt Shiva@localhost:/tmp
ssh: connect to host localhost port 22: Connection refused
rsync: connection unexpectedly closed (0 bytes received so far) [sender]
rsync error: unexplained error (code 255) at io.c(232) [sender=3.4.1]
Shiva@DESKTOP-MRCVEKT:~/practice12$
```

Screenshot 10: both fail with Connection refused.

Both said connect to host localhost port 22: Connection refused. This is the same thing telnet showed me before. scp and rsync both travel over ssh on port 22, and my computer is not running an ssh server. So the command is correct, but there is no server to receive the file. On a real server with ssh it would ask for the password and copy.

rsync and scp also work between two folders on the same computer, without any server. So I used that to see what the notes mean by rsync only sending the new files.

```
mkdir backup
rsync -av india/ backup/
touch india/agra.txt
rsync -av india/ backup/
scp small.txt backup/
ls backup
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice12'. The terminal shows the following commands and output:
1. `mkdir backup`
2. `rsync -av india/ backup/`
 Output: 'sending incremental file list', './', 'delhi.txt', 'karnataka/', 'karnataka/bengaluru.txt', 'karnataka/mysuru/', 'karnataka/mysuru/palace.txt', 'sent 338 bytes received 92 bytes 860.00 bytes/sec', 'total size is 0 speedup is 0.00'.
3. `touch india/agra.txt`
4. `rsync -av india/ backup/`
 Output: 'sending incremental file list', './', 'agra.txt', 'sent 277 bytes received 40 bytes 634.00 bytes/sec', 'total size is 0 speedup is 0.00'.
5. `scp small.txt backup/`
6. `ls backup`
 Output: 'agra.txt delhi.txt karnataka small.txt'.
7. The prompt `Shiva@DESKTOP-MRCVEKT:~/practice12$` is shown again.

Screenshot 11: the second rsync only sends agra.txt.

I used -av because plain rsync prints nothing. -a keeps the folders and file details, and -v shows the file names.

The first rsync copied everything: delhi.txt, the karnataka folder and the files inside. Then I made one new file, agra.txt, and ran the same rsync again. This time it only sent agra.txt, because everything else was already the same in backup. That is the incremental backup from the notes. scp does not check this, it copies the whole file every time.

Last, scp small.txt backup/ worked with no error, and ls backup shows small.txt next to the rsync files.

Summary of the network commands

Command	What it is for
ping	Check if another computer is alive and how fast it answers
telnet	Try to connect to one port on a computer, to see if it is open
netstat	See the network connections and open ports on my own computer
scp	Copy files safely between my computer and a server

Command	What it is for
rsync	Same job as scp, but only sends what is new or changed